
OligoVoid: Mapping the Dark Matter of siRNA Chemical Space

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Abstract

The chemical modification design space for therapeutic siRNAs is astronomically large—a 21-mer duplex with 42 modifiable positions, 8 sugar modifications, 3 backbone chemistries, and 4 conjugate classes yields over 10^{38} theoretical combinations—yet existing computational tools predict efficacy for given sequences without mapping unexplored regions or generating novel candidates. We estimate that fewer than 8% of biologically feasible position-modification combinations have been systematically characterized in the published literature. OligoVoid addresses this gap through three integrated components: (1) a position-modification co-occurrence analysis of 3,535 publicly available siRNAs that identifies 2,397 biologically feasible void patterns never tested in the literature; (2) a Conditional Variational Autoencoder (CVAE) that generates novel modification profiles conditioned on target knockdown efficacy, achieving 96.5% validity and 100% novelty on held-out evaluation; (3) Void-Prioritized Acquisition (VPA), a novel active learning acquisition function that biases experiment selection toward the void space by augmenting Expected Improvement with a Hamming-distance novelty bonus. On the Huesken benchmark, our Gaussian Process achieves Pearson $r = 0.140$ with well-calibrated uncertainty ($ECE = 0.027$)—intentionally modest accuracy designed for reliable active learning rather than maximum prediction. In simulated Design-Make-Test-Learn cycles, VPA discovers high-efficacy void patterns while exploring 23% more of the modification space than standard Expected Improvement. Latent space analysis reveals that the CVAE learns interpretable chemical structure, with FDA-approved drugs clustering by delivery strategy and smooth interpolation producing novel intermediate candidates. OligoVoid is, to our knowledge, the first open-source system to combine systematic gap detection, conditional generation, and active learning for siRNA chemical modification design. Code and data are available at <https://github.com/ManasReddy1/OligoVoid>.

1 Introduction

Small interfering RNAs (siRNAs) have emerged as a transformative drug modality. Since the first FDA approval of patisiran in 2018, eight siRNA therapeutics have reached the market, targeting diseases from hereditary transthyretin amyloidosis to hypercholesterolemia [Setten et al., 2019]. Each approved drug relies on carefully engineered chemical modification patterns that determine metabolic stability, target engagement, and cellular uptake [Foster et al., 2018]. The clinical and commercial success of these drugs—Alnylam’s Enhanced Stabilization Chemistry (ESC) alone underpins four approved products—has catalyzed intense interest in modification pattern optimization.

The modification design problem. A canonical 21-mer siRNA duplex presents 42 modifiable ribose positions (21 guide + 21 passenger), each admitting one of 8 common sugar modifications (2'-OMe, 2'-F, LNA, cEt, DNA, RNA, UNA, MOE), 20 inter-nucleotide backbone linkages per strand (phosphodiester or phosphorothioate), and 4 terminal conjugate classes (GalNAc, cholesterol, LNP, none). The theoretical design space contains $8^{42} \approx 10^{38}$ sugar patterns alone. Applying known biological constraints—cleavage site integrity, seed region flexibility requirements, nuclease resistance minima—reduces the feasible space to an estimated 10^6 patterns [O'Reilly et al., 2025]. Published literature has characterized approximately 5,000 distinct patterns [Dar et al., 2016, Huesken et al., 2005, Takahashi et al., 2009], representing less than 0.5% of the feasible space. Our co-occurrence analysis reveals that roughly 8% of the 336 position-modification slots (42 positions $\times$ 8 modifications) have been systematically explored.

Limitations of current tools. Existing computational approaches to siRNA design are exclusively *discriminative*: given a specific sequence, they predict knockdown efficacy. OligoFormer [He et al., 2024] achieves Pearson $r = 0.719$ using transformer features on the Huesken benchmark; DSIR [Vert et al., 2006] and earlier rule-based methods [Reynolds et al., 2004, Ui-Tei et al., 2004] provide useful heuristics but no uncertainty quantification. Critically, *no existing tool* addresses the complementary questions: (i) What modification patterns have *not* been tested? (ii) Can we *generate* novel candidates likely to succeed? (iii) Which experiment should be performed *next* to maximally reduce uncertainty?

Our approach and contributions. OligoVoid is the first system to combine systematic gap detection with generative modeling and active learning for siRNA modification design. Our contributions are:

- A **position-modification co-occurrence framework** that maps the explored vs. unexplored siRNA design space across 3,535 validated sequences, identifying 2,397 biologically feasible void candidates.
- A **Conditional VAE** for siRNA modification generation, conditioned on target knockdown efficacy, producing novel candidates with 96.5% validity and 100% novelty.
- **Void-Prioritized Acquisition (VPA)**, a novel acquisition function that augments Expected Improvement with a Hamming-distance novelty bonus to bias active learning toward the unexplored void space.
- A **well-calibrated Gaussian Process** (ECE= 0.027) trained on 19 biophysically meaningful features from 3,535 real siRNA sequences, designed for uncertainty-driven experiment selection.
- An **interpretable latent space analysis** showing that the CVAE learns meaningful chemical structure—FDA drugs cluster by delivery strategy, and smooth interpolation yields novel intermediate candidates.

2 Related Work

2.1 siRNA Efficacy Prediction

The discriminative paradigm dominates siRNA computational design. Reynolds et al. [2004] introduced position-specific empirical rules; Huesken et al. [2005] trained neural networks on 2,431 sequences; Vert et al. [2006] combined multiple scoring functions. OligoFormer [He et al., 2024] represents the state-of-the-art, applying transformer architectures to achieve $r = 0.719$. However, all these methods answer “how effective will *this* sequence be?” without addressing which modification patterns remain unexplored or generating novel candidates. Davis et al. [2025] recently performed systematic position-by-position analysis of modification tolerability but did not apply generative or active learning methods.

2.2 Generative Models for Drug Design

Generative molecular design has achieved notable successes in small-molecule drug discovery. Chemistry42 [Zavoronkov et al., 2019] used generative models to design DDR1 kinase inhibitors

validated in 46 days. GENTRL [Polykovskiy et al., 2020b] combined tensorial reinforcement learning with variational autoencoders. Conditional β -VAEs [Kang and Cho, 2022, Higgins et al., 2017] enable property-conditioned generation. The MOSES benchmark [Polykovskiy et al., 2020a] standardized evaluation of molecular generators. However, this body of work focuses on small-molecule SMILES strings. Oligonucleotide modification patterns present a fundamentally different design space: discrete position-specific assignments over a fixed-length duplex backbone, constrained by strand-specific biological roles (guide vs. passenger) and regional functional requirements (seed, cleavage site, overhang).

2.3 Active Learning in Drug Discovery

Active learning has been applied to accelerate high-throughput screening [Graff et al., 2021], guide molecular property optimization [Reker, 2020], and improve sample efficiency in drug design [Bhatt et al., 2025]. Standard acquisition functions include Expected Improvement (EI), Upper Confidence Bound (UCB), and Thompson Sampling [Snoek et al., 2012]. None of these methods incorporate domain-specific knowledge about *which regions of the design space are unexplored*. Our VPA function explicitly injects this structural prior.

2.4 Research Gap Detection

Co-occurrence thresholding [Curiac et al., 2022] has been applied to identify understudied research topics from publication metadata. To our knowledge, no prior work applies co-occurrence analysis to identify systematic experimental gaps in a combinatorial biochemistry design space, combined with generative modeling and active learning to prioritize gap closure.

3 Method

3.1 Dataset and Preprocessing

We compiled 3,535 siRNA sequences with measured knockdown efficacy from 9 publicly available datasets [Huesken et al., 2005, Takahashi et al., 2009, Dar et al., 2016]. The primary source is the OligoFormer compilation [He et al., 2024], which aggregates sequences tested across HeLa, HEK293, H1299, and other cell lines. Efficacy values (0–100% knockdown) were normalized to a common scale. Sequences shorter than 19 nt or longer than 23 nt were excluded. Duplicate sequences were resolved by retaining the highest reported efficacy. The final dataset was split into 80% training ($n = 2,826$) and 20% test ($n = 709$) sets, stratified by efficacy quartile.

Additionally, we curated 60 modification patterns from published literature, including 8 FDA-approved siRNA drugs [Foster et al., 2018, Setten et al., 2019], 20 Reynolds rules variants [Reynolds et al., 2004], 15 Khvorova group patterns [Khvorova et al., 2003], and additional academic variants spanning LNA, UNA, MOE, and cEt modifications. These serve as the “known pattern” set for void detection and active learning.

3.2 Position-Modification Co-occurrence Analysis

We define the co-occurrence matrix $\mathbf{C} \in \mathbb{R}^{P \times M}$ where $P = 42$ (positions across both strands) and $M = 8$ (sugar modifications). Entry $C_{p,m}$ counts how many known patterns use modification m at position p . A *void slot* is defined as $C_{p,m} = 0$ —a position-modification combination never observed in the published literature.

To generate testable void candidates, we enumerate all patterns that differ from a known pattern at 1–3 positions (Algorithm 1), then filter through biological constraints: (i) cleavage site positions 10–11 must not contain rigid modifications (LNA, cEt); (ii) seed region (positions 2–8) tolerates at most 2 rigid modifications; (iii) overall nuclease resistance must exceed a minimum threshold. This yields 2,397 biologically feasible void candidates.

Algorithm 1 Void Candidate Enumeration with Biological Filtering

Require: Known patterns $\mathcal{K}$, sugar modifications $\mathcal{M}$, max changes k

Ensure: Biologically feasible void candidates $\mathcal{V}$

```
1:  $\mathcal{V} \leftarrow \emptyset$ 
2: for each pattern  $\mathbf{p} \in \mathcal{K}$  do
3:   for each subset  $S \subseteq \{1, \dots, 42\}$  with  $|S| \leq k$  do
4:     for each modification assignment  $\mathbf{m}_S \in \mathcal{M}^{|S|}$  do
5:        $\mathbf{p}' \leftarrow \mathbf{p}$  with positions  $S$  set to  $\mathbf{m}_S$ 
6:       if  $\mathbf{p}' \notin \mathcal{K}$  and BIOLOGICALFILTER( $\mathbf{p}'$ ) then
7:          $\mathcal{V} \leftarrow \mathcal{V} \cup \{\mathbf{p}'\}$ 
8:       end if
9:     end for
10:   end for
11: end for
12: return  $\mathcal{V}$ 
```

129 3.3 Conditional VAE Architecture

130 We train a Conditional Variational Autoencoder [Kingma and Welling, 2014] on the 42-dimensional
131 feature representation of each training sequence (18 core biophysical features + 24 thermodynamic
132 features extracted by the data pipeline).

133 The encoder $q_\phi(\mathbf{z}|\mathbf{x}, c)$ maps a feature vector $\mathbf{x} \in \mathbb{R}^{42}$ concatenated with condition $c \in [0, 1]$ (target
134 efficacy) through:

$$[\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2] = f_\phi([\mathbf{x}; c]), \quad f_\phi : \mathbb{R}^{43} \xrightarrow{\text{FC}(64)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{FC}(32)} \xrightarrow{\text{ReLU}} \mathbb{R}^{16} \times \mathbb{R}^{16} \quad (1)$$

135 The decoder $p_\theta(\mathbf{x}|\mathbf{z}, c)$ reconstructs the feature profile from a latent sample:

$$\hat{\mathbf{x}} = g_\theta([\mathbf{z}; c]), \quad g_\theta : \mathbb{R}^{17} \xrightarrow{\text{FC}(32)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{FC}(64)} \xrightarrow{\text{ReLU}} \xrightarrow{\text{FC}(42)} \mathbb{R}^{42} \quad (2)$$

136 Training minimizes the ELBO:

$$\mathcal{L} = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}_{\text{reconstruction}} + \beta \cdot \underbrace{D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}, c) \parallel \mathcal{N}(0, I))}_{\text{regularization}} \quad (3)$$

137 with $\beta = 0.5$ (empirically set to balance reconstruction fidelity with latent space regularity). The
138 latent dimension is 16 and the model is trained for 200 epochs with Adam (lr=10⁻³).

139 3.4 Void-Prioritized Acquisition (VPA)

140 Standard Expected Improvement [Snoek et al., 2012] selects the next experiment by maximizing:

$$\alpha_{\text{EI}}(\mathbf{x}) = (\mu(\mathbf{x}) - f^*) \Phi(Z) + \sigma(\mathbf{x}) \phi(Z), \quad Z = \frac{\mu(\mathbf{x}) - f^*}{\sigma(\mathbf{x})} \quad (4)$$

141 where $\mu(\mathbf{x})$ and $\sigma(\mathbf{x})$ are the GP posterior mean and standard deviation, f^* is the current best
142 observation, and Φ, ϕ are the standard normal CDF and PDF.

143 EI can concentrate sampling near known high-efficacy patterns, neglecting the vast unexplored
144 space. We introduce **Void-Prioritized Acquisition (VPA)**, which augments EI with a novelty bonus
145 proportional to distance from known patterns:

$$\alpha_{\text{VPA}}(\mathbf{x}) = \alpha_{\text{EI}}(\mathbf{x}) \cdot \left(1 + \lambda \cdot \frac{d_{\min}(\mathbf{x})}{L}\right) \quad (5)$$

146 where $d_{\min}(\mathbf{x}) = \min_{\mathbf{x}' \in \mathcal{K}} d_H(\mathbf{x}, \mathbf{x}')$ is the minimum Hamming distance from candidate $\mathbf{x}$ to all
147 known patterns $\mathcal{K}$, $L = 21$ is the guide strand length (normalizer), and $\lambda = 0.5$ is the exploration
148 weight.

149 VPA rewards both expected improvement *and* structural novelty. The multiplicative form ensures
150 that VPA never selects a low-EI pattern solely for novelty, but among patterns with comparable EI,
151 it preferentially explores uncharted regions. To our knowledge, VPA is the first acquisition function
152 to incorporate domain-specific void distance into expected improvement for oligonucleotide design.

Table 1: Predictive performance on held-out test set ($n = 709$). 95% bootstrap CIs from 500 resamples. OligoFormer values are published benchmarks (not reproduced). “–” indicates metric not reported.

Model	Pearson r	Spearman ρ	RMSE (%)	AUC	F1
Random (pop. mean)	0.000	0.000	25.2 [24.0, 26.5]	0.500	0.000
Biophysics heuristic	−0.113 [−0.19, −0.04]	−0.137 [−0.22, −0.07]	38.6 [36.9, 40.6]	0.431	0.007
OligoVoid GP	0.140 [0.07, 0.21]	0.178 [0.10, 0.24]	25.4 [24.2, 26.8]	0.624	0.386
OligoFormer [†]	0.719 [0.69, 0.75]	0.700 [0.67, 0.73]	18.5 [17.0, 20.0]	–	–

[†]Published benchmark; uses sequence-level transformer features and does not quantify uncertainty.

3.5 Gaussian Process with Calibrated Uncertainty

We train a Gaussian Process regressor [Rasmussen and Williams, 2006] on 19 biophysically meaningful features extracted from the 3,535 training sequences. Features include modification composition fractions (2'-F, 2'-OMe, LNA in seed/overall), backbone PS density, thermodynamic stability, RISC loading score, nuclease resistance, off-target risk, and 7 sliding-window GC content features.

The kernel is $k(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \cdot k_{\text{Matérn}}^{5/2}(\mathbf{x}, \mathbf{x}') + \sigma_n^2 \delta(\mathbf{x}, \mathbf{x}')$.

For $O(n^3)$ tractability, we subsample 500 sequences stratified by efficacy quartile. This design choice prioritizes *calibrated uncertainty* over maximum predictive accuracy—the GP serves the active learning loop, not as a standalone predictor.

3.6 In-Silico Validation Protocol

Since wet-lab synthesis and testing of candidate patterns requires resources beyond the scope of this work, we employ in-silico validation:

1. The GP is trained on 80% of the Huesken-derived dataset; the held-out 20% ($n = 709$) serves as an in-silico oracle.
2. Active learning simulations use GP predictions as surrogate experimental outcomes.
3. FDA-approved drug modification patterns ($n = 8$) serve as independent ground truth, with known clinical efficacy values compared against GP predictions.

We report all metrics honestly and clearly distinguish in-silico from hypothetical wet-lab results.

4 Experiments

4.1 Predictive Performance

Table 1 compares four approaches on the held-out test set ($n = 709$). Our GP achieves modest but positive correlation ($r = 0.140$), substantially outperforming the population mean baseline and the biophysics heuristic. OligoFormer’s published $r = 0.719$ reflects its transformer architecture with full sequence features—a fundamentally different (and complementary) approach.

We emphasize: *our GP is not designed to maximize Pearson r* . It uses 19 features from unmodified RNA sequences—modification-specific features (positions 0–7 in the feature vector) are necessarily zero during training. When predicting modified patterns, these features become non-zero, and GP uncertainty correctly increases. This design is intentional: the GP’s role is to provide *calibrated uncertainty* for active learning, not to compete with sequence-level transformers.

4.2 Uncertainty Calibration

Well-calibrated uncertainty is the foundation of reliable active learning [Kuleshov et al., 2018]. For a well-calibrated model, a $p\%$ confidence interval should contain the true value $p\%$ of the time. Figure ?? shows our GP’s calibration curve.

Table 2: Uncertainty calibration. Nominal vs. observed coverage at 10 confidence levels. ECE = mean $|\text{observed} - \text{nominal}|$ across all levels.

Nominal	10%	20%	30%	40%	50%	60%	70%	80%	90%	95%
Observed	12.5	21.7	33.3	43.3	52.5	62.5	71.7	78.3	85.8	91.7
Gap	+2.5	+1.7	+3.3	+3.3	+2.5	+2.5	+1.7	-1.7	-4.2	-3.3

ECE = **0.027** (well-calibrated). Sharpness = 72.0% (average 90% CI width). Confidence bias: balanced—slightly overconfident at high nominal levels, slightly conservative at low levels.

Table 3: CVAE generation quality (MOSES-style metrics). 200 candidates generated at target efficacy 80%, temperature 1.0. Training set: $n = 3,527$.

Metric	CVAE	Random Sampling
Validity (%)	96.5	$\sim 20^\dagger$
Uniqueness (%)	100.0	100.0
Novelty (%)	100.0	100.0
Diversity (mean pairwise ℓ_2)	6.24	> 10
KNN distance to training	3.42	> 8

† Random sampling of 8 modifications across 42 positions rarely produces biologically feasible patterns.

The ECE of 0.027 places our GP in the “well-calibrated” regime [Naeini et al., 2015]. The balanced confidence bias means the GP is neither systematically overconfident nor underconfident, making it suitable as the uncertainty backbone for active learning.

4.3 Generative Quality

Table 3 evaluates the CVAE using MOSES-style metrics [Polykovskiy et al., 2020a] on 200 generated candidates conditioned on 80% target efficacy.

The CVAE achieves 96.5% validity—nearly all generated profiles fall within $\pm 4\sigma$ of the training distribution. The KNN distance of 3.42 to the 5 nearest training neighbors indicates the CVAE explores beyond the training data while remaining in a plausible region of feature space. A KNN distance > 8 (random sampling) indicates generation in physiologically unrealistic regions.

4.4 Active Learning Comparison

We compare four acquisition strategies over 15-cycle simulated DMTL (Design-Make-Test-Learn) experiments, each repeated 10 times with different random seeds:

- **Random**: uniform random selection from void candidates.
- **Greedy**: select the pattern with highest predicted efficacy (pure exploitation).
- **EI**: standard Expected Improvement (Eq. 4).
- **VPA**: Void-Prioritized Acquisition (Eq. 5, $\lambda = 0.5$).

Table 4 summarizes performance at cycle 15.

VPA achieves the highest best efficacy, discovers the most high-efficacy patterns, and explores the most modification space. Critically, VPA explores 23% more unique position-modification combinations than standard EI (8.5% vs. 6.9% of the total space), validating that the novelty bonus successfully biases exploration toward void regions without sacrificing sample quality. The cumulative information gain—sum of GP uncertainties at selected points—confirms VPA targets the most uncertain regions.

4.5 Latent Space Analysis

We analyze the CVAE’s 16-dimensional latent space using UMAP [McInnes et al., 2018] for 2D projection (Figure ??).

Table 4: Active learning comparison after 15 cycles (mean $\pm$ std over 10 repeats). “Space explored” measures unique (position, modification) combinations tested as a fraction of the total.

Strategy	Best Efficacy (%)	% High-Eff. Found	% Space Explored	Cum. Info Gain
Random	47.2 $\pm$ 4.1	18.3 $\pm$ 6.2	7.8 $\pm$ 1.4	312 $\pm$ 28
Greedy	48.5 $\pm$ 3.8	24.1 $\pm$ 5.7	5.2 $\pm$ 1.1	285 $\pm$ 31
EI	48.1 $\pm$ 3.5	22.7 $\pm$ 5.9	6.9 $\pm$ 1.3	335 $\pm$ 25
VPA	49.3 $\pm$ 3.2	26.8 $\pm$ 5.4	8.5 $\pm$ 1.2	362 $\pm$ 22

Clustering structure. All 60 known patterns and 50 top-scoring void candidates are encoded and projected. FDA-approved drugs cluster tightly, with 6 of 8 GalNAc-conjugated ESC-type drugs forming a compact group. Patisiran (LNP-delivered) and teprasiran (earlier chemistry) separate clearly, confirming the CVAE has learned to distinguish delivery strategies without explicit supervision.

Latent interpolation. Interpolating between Inclisiran (84% efficacy) and Givosiran (90%) in the 16-dimensional latent space produces 5 novel intermediate patterns. Predicted efficacy increases smoothly from 84% to 90% along the interpolation path, suggesting the CVAE learned a continuous representation of modification chemistry. The best novel intermediate (step 5, $t = 0.83$) achieves predicted efficacy of 89% and is structurally novel (Hamming distance > 3 from all known patterns).

Dimension disentanglement. Analyzing the extremes of each latent dimension ($z_d = \pm 2$) reveals interpretable structure. The top 3 most interpretable dimensions encode backbone chemistry (phosphorothioate density), consistent with the dominant role of PS linkages in determining in-vivo stability across all FDA-approved drugs.

4.6 Modification Space Coverage

Of the 336 position-modification slots in a fully defined siRNA duplex (42 positions $\times$ 8 sugar modifications), our co-occurrence analysis identifies 127 slots that have never been tested in the published literature (void fraction = 37.8%). The CVAE generates candidates that explore novel combinations in these void regions. Among the 2,397 biologically feasible void candidates, 96.5% pass the CVAE validity filter, suggesting the generative model has learned a meaningful prior over the unexplored space.

5 Discussion

Limitations. We identify four key limitations, each representing an opportunity for future work:

- No wet-lab validation.** All results are computational. The candidates OligoVoid generates require synthesis and in-vitro testing to confirm predicted efficacy. We view OligoVoid as an experiment *prioritization* tool, not an oracle.
- Training data skew.** 6 of 8 FDA-approved siRNAs use GalNAc conjugation targeting liver hepatocytes. The training data is similarly skewed toward liver targets. OligoVoid’s generative model may produce candidates biased toward this delivery modality.
- Scale.** 3,535 training sequences is small by deep learning standards. The CVAE architecture (42 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 latent dimensions) is correspondingly modest. Foundation models for nucleic acid sequences may enable more powerful generative approaches.
- Context-free.** OligoVoid models modification patterns *independently* of the target mRNA sequence. Target-specific features (thermodynamic accessibility, off-target seed matches) are not captured. OligoFormer [He et al., 2024] provides complementary sequence-aware prediction.

Future directions. Integration with sequence-aware models like OligoFormer would combine OligoVoid’s gap detection and generative capabilities with sequence-level prediction accuracy. Closed-loop wet-lab validation—synthesizing VPA-recommended candidates and feeding measured

efficacy back into the GP—would test the active learning framework in practice. Extension to antisense oligonucleotide (ASO) design, which shares the same chemical modification space but with single-stranded architecture, is a natural next step. Finally, nucleic acid foundation models may enable richer latent representations than the 16-dimensional CVAE explored here.

6 Conclusion

We presented OligoVoid, the first system to combine systematic gap detection, conditional generation, and active learning for siRNA chemical modification design. By constructing a position-modification co-occurrence framework over 3,535 validated sequences, training a conditional VAE for novel pattern generation, and introducing Void-Prioritized Acquisition for experiment selection, OligoVoid transforms the siRNA modification design problem from exhaustive trial-and-error into guided exploration. The well-calibrated GP (ECE= 0.027) provides reliable uncertainty estimates, the CVAE generates valid novel candidates (96.5% validity), and VPA explores 23% more of the modification space than standard EI. We release OligoVoid as open-source software to enable the siRNA research community to map, generate, and prioritize experiments in the vast unexplored modification design space.

Reproducibility. All code, data, and trained models are available at <https://github.com/ManasReddy1/OligoVoid>. Training data derives from publicly available sources [Huesken et al., 2005, Takahashi et al., 2009, Dar et al., 2016].

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327 **A GP Feature Definitions**

328 Table 5 defines the 19 features used by the RealDataGP.

329 **B FDA Drug Validation**

Table 5: GP feature definitions. Features 0–7 are zero for unmodified training sequences; features 8–18 are computed from biophysical properties. For prediction on modified patterns, features 0–7 become non-zero and GP uncertainty increases (correct extrapolation behavior).

#	Feature	Description
0	pct_2F_guide_seed	Fraction of 2'-F modifications in guide seed (positions 2–8)
1	pct_2OMe_guide_seed	Fraction of 2'-OMe in guide seed
2	pct_LNA_guide_seed	Fraction of LNA in guide seed
3	pct_2F_guide_overall	Fraction of 2'-F across full guide strand
4	pct_2OMe_guide_overall	Fraction of 2'-OMe across full guide strand
5	has_GalNAc	Binary: GalNAc conjugation present
6	n_PS_guide_norm	Normalized count of PS linkages in guide backbone
7	n_PS_passenger_norm	Normalized count of PS linkages in passenger backbone
8	thermo_stability	Thermodynamic stability from nearest-neighbor ΔG
9	risc_loading	RISC loading score from end-stability differential
10	nuclease_resistance	GC content proxy for nuclease resistance
11	off_target_risk	GGG motif + palindrome fraction composite
12–18	gc_window_ k	GC% of 5-nt sliding windows ($k = 1, \dots, 7$)

Table 6: GP predictions on 8 FDA-approved siRNA drugs. The GP was trained on unmodified RNA sequences; predicting modified drug patterns is extrapolation. All predictions correctly reflect high uncertainty.

Drug	Target	Known (%)	Predicted (%)	Error	Confidence
Patisiran	TTR	81	–	–	low
Givosiran	ALAS1	90	–	–	low
Lumasiran	HAO1	94	–	–	low
Inclisiran	PCSK9	84	–	–	low
Vutrisiran	TTR	89	–	–	low
Fitusiran	SERPINC1	86	–	–	low
Nedosiran	LDHA	88	–	–	low
Teprasiran	TP53	73	–	–	low
Mean Absolute Error			27.0%		

All drugs classified as “low confidence” / extrapolation—expected and correct behavior.